

NeoCortex: A Neuromorphic Virtual Drug Trial Platform with Emergent Neurochemical-Physiological Coupling

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Abstract

Drug development for central nervous system (CNS) disorders faces a 90% Phase II/III failure rate, costing \$100-200M per failure. We present NeoCortex, a neuromorphic platform simulating up to 100,000 spiking neurons (scalable to millions on GPU, with neuromorphic hardware support via Intel Loihi) with complete neuromodulatory systems (6 neurotransmitters, 13 receptors) and advanced plasticity mechanisms. The system derives physiological readouts from simulated neurochemistry through neurochemical-physiological coupling, without outcome rules written into the model. Retrospective validation shows directional agreement with published clinical outcomes across three pathologies (Depression, Epilepsy, Parkinson's). A key validation study comparing Fluoxetine (FDA-approved 1987) versus Reboxetine (FDA-rejected 2001) shows correct discrimination: NeoCortex reproduces Fluoxetine efficacy (5HT +47.7%, fatigue from 60% to 51% over 48 hours) and Reboxetine failure (5HT +2.5%, insufficient symptom relief). Additionally, the platform's integrated Drug Discovery module generated NEURO-ACh v3.0-OH, a candidate dual-target compound (AChE + SERT inhibition) hypothesised to offer better gastrointestinal tolerability than existing treatments; it has not yet been synthesised or tested. NeoCortex is a research platform for virtual clinical trials in CNS drug development, and prospective validation is the next step.

Keywords: Virtual drug trials, spiking neural networks, neuromodulation, synaptic plasticity, drug discovery, computational neuroscience, pharmacology, Alzheimer's disease

1. Introduction

1.1 The CNS drug development crisis

Central nervous system drug development faces unprecedented challenges. Current failure rates exceed 90% in Phase II/III clinical trials, with average costs of \$100-200 million per failed candidate and 2-3 years of lost development time. Traditional approaches — molecular docking, static models, and animal studies — fail to predict emergent physiological effects arising from dynamic interactions across multiple neurotransmitter systems.

The fundamental limitation is the inability to simulate the complex, non-linear dynamics of integrated brain systems under pharmacological perturbation. A drug targeting serotonin reuptake does not simply increase serotonin levels: it triggers cascading changes across dopaminergic, noradrenergic, and GABAergic circuits, with emergent effects on neuroplasticity, receptor expression, and ultimately observable physiology.

Two structural obstacles compound the problem. The functional complexity of the brain, combined with the blood-brain barrier, drastically limits bioavailability: over 98% of potentially active molecules fail to reach effective concentrations in brain parenchyma without specialised transport systems. Although engineered nanosystems, receptor-mediated transport and focused ultrasound have been proposed, the barrier remains a primary obstacle for safety, scalability and predictivity reasons. Concurrently, several severe neurological diseases remain without truly effective treatments — Alzheimer's disease, glioblastoma, amyotrophic lateral sclerosis — where clinical heterogeneity, the lack of reliable biomarkers and the difficulty of measuring therapeutic response contribute to failure in advanced trial phases.

1.2 The NeoCortex solution

NeoCortex addresses this gap through a neuromorphic simulation platform modelling 100,000 biologically realistic spiking neurons and approximately 10 million synapses, organised into cortical columns and subcortical

structures, GPU-scalable to 1M+ neurons with Intel Loihi integration enabling real-time simulation at larger scales. The system implements six complete neuromodulatory systems (serotonin, dopamine, noradrenaline, acetylcholine, GABA, glutamate) with 13 receptor subtypes, four advanced plasticity mechanisms (BCM, synaptic scaling, three-factor learning, short-term plasticity), and eight physiological parameters computed from the simulated neurochemical dynamics without pre-programmed outcomes.

Unlike traditional deterministic models, NeoCortex generates emergent behaviours derived from the dynamic interaction between pathological neurochemical state, molecular mechanism of drug action, pharmacokinetics and pharmacodynamics, synaptic plasticity and adaptation, and the coupling between neurotransmitters and global physiological parameters.

In particular, NeoCortex offers: (i) more accurate preclinical screening, through detailed simulation of neurochemical and receptor mechanisms; (ii) dose optimisation before clinical trials, with the ability to test thousands of combinations in a few hours; (iii) an estimated 40% reduction in preclinical costs and development time from years to weeks; and (iv) the possibility of conducting virtual clinical trials, including personalised digital twins and individualised therapeutic response simulations, with potential regulatory applications.

2. System Architecture

2.1 Four-layer emergent architecture

NeoCortex employs a hierarchical architecture in which complex behaviours emerge from biologically constrained interactions without explicit rules:

Layer	Components	Function
1. Neural core	100K+ spiking neurons (GPU: 1M+ scalable), 4 columns x 6 layers	Leaky Integrate-and-Fire dynamics, spike propagation, realistic neural computation
2. Neuromodulation	6 neurotransmitters, 13 receptor subtypes	NT concentration dynamics, receptor activation (Hill equation), synaptic modulation, release-reuptake dynamics
3. Plasticity	BCM, synaptic scaling, three-factor learning, short-term plasticity	Adaptive weight changes, homeostatic regulation, reward-gated learning
4. Physiology	8 emergent parameters, 48 NT-coupling coefficients	Fatigue, muscle tension, pain, heart rate and others, computed from NT state

2.2 Neuromodulatory system

Each neurotransmitter follows biophysically realistic dynamics:

$$C(t+1) = C(t) + (Release - Reuptake - Decay) \times dt$$

where Release depends on presynaptic activity, Reuptake is modulated by transporter expression (the drug target), and Decay follows first-order kinetics. Receptor activation follows the Hill equation with neurotransmitter-specific EC50 and Hill coefficients derived from literature.

NT	Primary function	Receptors	Clinical relevance
Serotonin (5HT)	Mood, sleep, appetite	5HT1A, 5HT2A, 5HT2C	Depression, anxiety (SSRI target)
Dopamine (DA)	Reward, movement	D1, D2, D3	Parkinson's, schizophrenia
Noradrenaline (NE)	Attention, arousal	α 1, α 2, β	ADHD, depression (SNRI target)
Acetylcholine (ACh)	Memory, learning	M1, M2, nicotinic	Alzheimer's (AChE inhibitors)
GABA	Inhibition	GABA-A, GABA-B	Epilepsy, anxiety (benzodiazepines)
Glutamate	Excitation	NMDA, AMPA, kainate	Epilepsy, neurotoxicity

2.3 Emergent neurochemical-physiological coupling

A critical innovation is the system of universal formulas linking neurotransmitter states to eight observable physiological parameters. These formulas were derived empirically from clinical literature and represent a quantitative model of NT-physiology coupling applicable across multiple pathologies. Representative examples:

$$\text{fatigue} = 0.30 + (5\text{HT_deficit} \times 0.30 + \text{DA_deficit} \times 0.25 + \text{NE_deficit} \times 0.20 + \text{ACh_deficit} \times 0.10 + \text{GABA_deficit} \times 0.08 + \text{Glu_excess} \times 0.07) \times 0.55$$

$$\text{muscle_tension} = 0.35 + (\text{EI_tension} \times 0.40 + \text{NE_tension} \times 0.35 + \text{DA_rigidity} \times 0.25) \times 0.40$$

$$\text{pain} = \max(0, \min(0.30, 5\text{HT_deficit} \times 0.20 + \text{NE_deficit} \times 0.10 + \text{EI_imbalance} \times 0.05))$$

Each formula combines up to six neurotransmitter terms with a scale factor; across the eight parameters the full 8 × 6 matrix holds 48 coupling coefficients, some of them zero, representing literature-derived contributions of each neurotransmitter to the physiological output. These formulas operate on the actual simulated NT concentrations — not on pre-programmed values — so outcomes follow from the simulated dynamics, while the coefficients themselves encode weights taken from the clinical literature.

Parameter	5HT	DA	NE	ACh	GABA	Glu	Scale
Fatigue	0.30	0.25	0.20	0.10	0.08	0.07	0.55
Muscle tension	—	0.25	0.35	—	-0.20	0.40	0.40
Pain	0.20	—	0.10	—	-0.05	0.05	cap
Heart rate	0.10	0.35	—	—	0.20	-0.35	-0.10
Respiratory	0.10	—	0.35	-0.20	-0.25	0.25	—
Skin conductance	0.10	0.30	0.40	0.05	-0.10	0.15	0.45
Gut feeling	0.50	0.10	0.10	0.10	0.10	0.10	0.35
Temperature	0.20	—	±0.25	0.35	-0.15	—	—

Negative coefficients indicate inverse relationships (e.g. high 5HT reduces heart rate). Em dash indicates zero contribution.

2.4 Advanced plasticity mechanisms

The system implements four biologically validated plasticity mechanisms that maintain network stability during pharmacological perturbations, a requirement absent from previous models. All four are fully integrated and active at every simulation step, wrapped around every neuron group (pyramidal, PV, SST, VIP, chandelier) at trial initialisation.

Mechanism	Timescale	Function	Clinical relevance
BCM rule	Minutes-hours	Sliding threshold prevents runaway excitation	Prevents instability during chronic SSRI treatment
Synaptic scaling	Hours-days	Homeostatic normalisation of synaptic weights	Maintains stable firing rates over treatment
Three-factor learning	Seconds-minutes	Dopamine-gated synaptic modification	Critical for Parkinson's treatment simulation
Short-term plasticity	Milliseconds-seconds	Frequency-dependent facilitation/depression	Realistic temporal filtering of signals

BCM maintains an adaptive threshold $\theta(t+1) = \theta(t) + \alpha\theta \times (r^2(t) - \theta(t)) \times dt$, with weight change $\Delta w = \eta \times r_{\text{pre}} \times r_{\text{post}} \times (r_{\text{post}} - \theta) \times dt$. The threshold timescale is 10,000 ms, the learning rate 0.01, and each neuron carries an independent θ updated at every 1 ms step. During chronic SSRI treatment, elevated serotonin raises excitability; without BCM this drives firing rates above 15 Hz and eventual numerical failure, whereas BCM holds a stable 5-7 Hz while preserving the therapeutic neurochemical change.

Synaptic scaling normalises all synapses onto a neuron toward a 5 Hz target every 1,000 steps at rate 0.1, providing coarse-grained homeostasis that complements BCM's per-synapse adjustment. Turrigiano and Nelson (2004) demonstrated exactly this behaviour in cortical neurons over hours-to-days timescales.

Three-factor learning gates plasticity on neuromodulatory signal. An eligibility trace $e(t)$ captures recent pre-post correlations with a 1,000 ms timescale, and weight change follows $\Delta w = \eta \times e(t) \times DA(t)$, requiring DA above 0.5 to mimic D1 receptor activation. It is primarily active in striatal circuits. In Parkinson's simulations the mechanism is severely impaired at a 5.7% dopamine baseline, and levodopa only partially restores it — matching clinical observations of modest motor learning improvement in advanced disease. Reynolds and Wickens (2002) showed that striatal plasticity requires coincident dopamine release, glutamatergic input and postsynaptic depolarisation: the three factors implemented here.

Short-term plasticity provides millisecond-to-second dynamics through activity-dependent depression and facilitation, with effective efficacy $w_{\text{eff}} = w \times D \times (1 + F)$. Synapse-type parameters follow Tsodyks and Markram (1997): excitatory $U = 0.5$, $\tau_D = 1,100$ ms, $\tau_F = 50$ ms; inhibitory $U = 0.25$, $\tau_D = 700$ ms, $\tau_F = 20$ ms; thalamocortical $U = 0.15$, $\tau_D = 800$ ms, $\tau_F = 10$ ms. Functionally it acts as a high-pass filter, reducing efficacy under repetitive stimulation above 10 Hz while enhancing transmission during burst firing.

Simulations without plasticity exhibit network instability — runaway excitation and numerical failure — and do not produce usable trajectories. With full plasticity integration, firing rates remain stable throughout the 48-hour protocol and the neurochemical changes can be compared with clinical data.

2.5 Hardware implementation and scalability

The baseline implementation uses NVIDIA A100 GPUs with PyTorch-based neural simulation. Connectivity matrices are stored in Compressed Sparse Row format: with approximately 10M synapses (about 100 per neuron, roughly 0.1% connectivity), storage falls from 40GB for a dense FP32 $100K \times 100K$ matrix to about 80MB, and sparse matrix-vector multiplication achieves 50-100x speedup. Mixed precision retains FP32 only for accumulator variables, cutting memory bandwidth by 40% while holding numerical error below 1%.

Parallelisation is hierarchical. Each neuron is assigned an independent GPU thread for membrane potential integration. Synaptic current computation is parallelised across 10M synapses using sparse CSR with atomic summation. The four cortical columns are simulated independently for local computation, with 80% of connections intra-column and 20% inter-column. Multi-GPU scaling is under development, distributing neurons or partitioning columns across devices, targeting 1M+ neurons with under 5% communication overhead via NVLink.

Measured performance at 100,000 neurons and 10M synapses: latency of 10-15 ms per simulation step, equivalent to about 10M neuron updates per second, and 45-60 seconds total for a 48-hour virtual trial (2,000 warmup steps plus 2,880 treatment steps) at 300W sustained, or about 4-5 Wh of energy. Scaling remains near-linear to 500K neurons (42 ms/step) and has been demonstrated at 1M neurons (95 ms/step).

On Intel Loihi, floating-point values are converted to int32 with 12-bit fractional precision, holding quantisation error below 0.1% for neurotransmitter concentrations. Computation is event-driven, so typical cortical sparsity of 5-10% active neurons per step yields 10-100x efficiency gains, and all four plasticity mechanisms run on-chip through the programmable learning engine, eliminating host communication.

Metric	GPU A100	Loihi (single chip)	Loihi (10 chips)
Simulation time	45-60 sec	2-3 min	30-40 sec
Energy (48h trial)	3.8-5.0 Wh	0.03-0.05 Wh	0.08-0.11 Wh
Energy efficiency	1x	~100x	~45x
Latency per step	10-15 ms	<1 ms	<1 ms
Max neurons	1M+	100K	1M+
Hardware access	\$15,000	Research access (Intel INRC)	Research access (Intel INRC)

Energy is power multiplied by wall-clock simulation time, assuming 300 W for the A100 and about 1 W per Loihi chip. Loihi chips are not sold commercially; access is through Intel's Neuromorphic Research Community.

For drug discovery, a genetic algorithm evaluating 20 candidates sequentially takes about 10 minutes; batched on GPU this falls to roughly 40 seconds, and on a 20-chip Loihi network to about 2 minutes with roughly 5x lower energy consumption than the batched GPU run. Loihi's 1W power budget and sub-millisecond per-step latency make patient-specific simulation on edge devices a plausible route to point-of-care clinical decision support.

3. Methods

3.1 Disease modelling

Disease states are implemented through neurotransmitter efficacy reductions that propagate throughout the network. For Major Depressive Disorder: 5HT efficacy 0.60 (40% deficit), DA efficacy 0.70 (30% deficit), NE efficacy 0.75 (25% deficit). These values establish baseline neurochemical states and corresponding physiological parameters via the coupling formulas.

3.2 Pharmacological modelling

Drug mechanisms are implemented at the molecular level through transporter, enzyme and receptor interactions. Fluoxetine, for example, is entered as 85% SERT inhibition at therapeutic plasma concentration (C_{max} around 300 ng/mL), producing reuptake blockade that accumulates serotonin over the simulation period. Pharmacokinetics follows a one-compartment model with absorption and elimination:

$$C(t) = (D/V) \times (ka/(ka - ke)) \times (exp(-ke \times t) - exp(-ka \times t))$$

where D is dose, V volume of distribution, ka the absorption rate constant and ke the elimination rate constant. Drug-specific half-life and bioavailability are derived from published pharmacokinetic studies.

3.2.1 Hill equation pharmacodynamics

Drug-target interactions follow the Hill equation, modelling cooperative binding kinetics:

$$Occupancy = [Drug]^n / (EC_{50}^n + [Drug]^n)$$

where EC₅₀ is the concentration producing 50% occupancy and n the Hill coefficient. Coefficients above 1 indicate positive cooperativity, where initial binding facilitates subsequent events; values of 1.2-1.8 reflect the multi-subunit structure of most transporters and receptors, and directly determine the steepness of dose-response curves.

Target category	Specific target	Hill coefficient
Reuptake inhibitors	SERT / NET / DAT / GAT inhibition	1.3 - 1.7
Enzyme inhibitors	AChe, MAO-A, MAO-B, GABA transaminase	1.5 - 1.6
Receptor antagonists / agonists	D2 antagonism / agonism / partial agonism	1.4 - 1.7
	5-HT _{2A} antagonism, 5-HT _{1A} agonism	1.5 - 1.6
	Alpha-2 antagonism, H1 antagonism	1.4 - 1.5

GABA modulators	GABA-A positive allosteric modulation	1.6 - 1.8
Glutamate modulators	NMDA antagonism, glutamate release inhibition	1.5
Ion channels	Voltage-gated Na ⁺ , α 2delta Ca ²⁺ blockade	1.5

3.2.2 Active metabolite modelling

Many CNS drugs undergo hepatic metabolism producing pharmacologically active metabolites that contribute significantly to clinical efficacy and duration of action. NeoCortex models metabolite formation, pharmacokinetics and target interactions explicitly. Metabolite pharmacodynamics are modelled identically to parent drugs, with independent concentration profiles governed by formation kinetics, metabolite-specific half-lives and time-to-peak concentrations. Combined parent and metabolite activity determines total pharmacological effect.

Parent drug	Active metabolite	Formation	t _{1/2} (h)	Primary target(s)	Potency (Hill)
Fluoxetine	Norfluoxetine	80%	336	SERT inhibition	0.75 (1.5)
Sertraline	Desmethylsertraline	60%	62	SERT inhibition	0.15 (1.4)
Venlafaxine	O-desmethylvenlafaxine	90%	11	SERT/NET inhibition	0.85/0.85 (1.5)
Risperidone	9-Hydroxyrisperidone	85%	24	D2/5-HT2A antagonism	0.85/0.95 (1.7/1.6)
Aripiprazole	Dehydroaripiprazole	75%	94	D2 partial / 5-HT2A antag.	0.30/0.36 (1.4)
Quetiapine	Norquetiapine	50%	12	NET inhibition	0.15 (1.2)
Bupropion	Hydroxybupropion	70%	20	DAT/NET inhibition	0.35/0.30 (1.3/1.2)
Mirtazapine	Desmethylmirtazapine	50%	40	Alpha-2 antagonism	0.30 (1.5)
Diazepam	Nordiazepam	70%	100	GABA-A PAM	0.75 (1.6)
Alprazolam	Alpha-hydroxyalprazolam	40%	12	GABA-A PAM	0.45 (1.5)

Fluoxetine's norfluoxetine metabolite has a 14-day half-life and contributes significantly to sustained antidepressant effects even after parent drug washout. Further drugs are modelled without active metabolites, including paroxetine, escitalopram, duloxetine, methylphenidate, donepezil, levodopa and lorazepam.

Hill equations and active metabolite modelling bring the simulated pharmacodynamics closer to published dose-response and exposure data. Expected residual sources of discrepancy include drug-drug interactions through CYP450 competition, genetic polymorphisms such as CYP2D6 and COMT variants, and minor metabolites contributing under 10% of activity; none of these is modelled yet.

These enhancements also give the platform the mechanism behind fluoxetine's 4-6 week delay to full therapeutic effect, namely norfluoxetine accumulation. Reproducing that full time course requires simulations longer than the 48-hour window validated here (see Section 6.4).

3.3 Simulation protocol

Each trial follows a standardised protocol: (1) network initialisation with 100,000 neurons across cortical columns and subcortical structures; (2) disease state application establishing neurochemical deficits; (3) a warmup period of 2,000 simulation steps for network stabilisation; (4) baseline measurement before treatment; (5) drug administration with pharmacokinetic profile over 48 hours at 60 steps per simulated hour; (6) continuous monitoring of neurotransmitter concentrations and physiological parameters.

4. Validation Studies

To validate predictive accuracy across multiple pathologies and pharmacological mechanisms, we conducted three independent retrospective validation studies: a comparison of Fluoxetine (FDA-approved 1987) versus Reboxetine (FDA-rejected 2001) for Major Depressive Disorder; Lamotrigine for epilepsy; and Levodopa for

Parkinson's disease. These cases span different neurotransmitter systems and disease mechanisms, providing a test of the NT-physiology coupling framework across mechanisms.

4.1 Data provenance and simulation parameters

The values reported throughout this section were obtained from a series of independent simulation runs carried out during platform development. Each run was executed at the standard therapeutic dose for the compound under test, against the corresponding disease model, over a 48-hour window with sampling every 24 hours. The parameters used were:

Study	Compound	Dose	Primary target(s)	Disease model
1a	Fluoxetine	20 mg (range 10-80)	SERT inhibition 0.85; norfluoxetine 0.75	MDD: 5HT -40%, DA -30%, NE -25%
1b	Reboxetine	8 mg (range 4-12)	NET inhibition 0.90	MDD: same profile
2	Lamotrigine	200 mg (range 25-400)	Na ⁺ channel blockade 0.70; glutamate release inhibition 0.50	Epilepsy: Glu 65%, GABA 36%
3	Levodopa	100 mg (range 100-1000)	DA precursor 0.90	Advanced PD: DA 5.7%

The model is stochastic: sensory input during warmup and treatment is drawn at random and no fixed seed is set, so absolute values vary between runs of the same configuration. All figures in this paper are taken from the runs reported in Sections 4.2.3, 4.3.3 and 4.4.3. Relative changes and the direction of effect are stable across runs; individual absolute concentrations are not intended as exact reproducible values.

4.2 Study 1: Major Depressive Disorder — Fluoxetine versus Reboxetine

4.2.1 Pathophysiology

Major Depressive Disorder is characterised by persistent low mood, anhedonia and cognitive impairment arising from dysregulation of monoaminergic systems. It is not the simple reduction of a single neurotransmitter that determines the disorder, but a complex dysfunction of release, reuptake and receptor regulation within critical circuits. Alongside monoamines, glutamate, GABA and acetylcholine also contribute, confirming the heterogeneous and multisystemic nature of the disorder.

Dopamine plays a central role in reward, motivation and behavioural initiative; reduced synthesis or release in mesolimbic, prefrontal and limbic regions is associated with anhedonia, apathy and reduced motivation. Noradrenaline regulates arousal, attention, stress response and mood tone. Serotonin, beyond mood regulation, modulates sleep, appetite, gastrointestinal motility and sexual function. Neuroimaging studies demonstrate reduced serotonin transporter binding and altered 5-HT_{1A} receptor density in prefrontal cortex and amygdala.

Currently available antidepressants increase synaptic monoamine availability through distinct mechanisms: SSRIs through selective SERT inhibition; SNRIs through combined SERT and NET blockade; tricyclics through non-selective inhibition with significant receptor side effects; NDRIs through dopamine and noradrenaline reuptake inhibition; and MAOIs through degradation inhibition.

4.2.2 Reboxetine background and FDA rejection

Reboxetine is the first commercially available selective noradrenaline reuptake inhibitor, developed specifically as a first-line therapy for major depression. It increases NE availability at synapses, promoting improvements in energy, attention and motivation, with poor affinity for serotonergic, histaminergic and cholinergic receptors and a tolerability profile more favourable than tricyclics. However, comprehensive meta-analyses revealed it inferior to SSRIs in remission and response rates. A landmark 2010 study (Eyding et al., BMJ) analysing both published and unpublished placebo-controlled and SSRI-controlled trials found reboxetine overall ineffective for major depressive disorder. The FDA rejected it in 2001, citing lack of substantial evidence of effectiveness.

Fluoxetine, developed by Eli Lilly as the first highly selective SSRI, acts almost exclusively on the serotonin transporter. Its introduction represented a turning point in modern psychopharmacology. The two compounds constitute an ideal test case: both treat depression through monoamine reuptake inhibition but differ fundamentally in selectivity and clinical efficacy.

Parameter	Fluoxetine	Reboxetine
Mechanism	Selective serotonin reuptake inhibitor	Selective noradrenaline reuptake inhibitor
Primary target	SERT (85% inhibition)	NET (90% inhibition)
Off-targets	NET <5%, DAT <5%	SERT 5%, DAT <5%
FDA status	Approved 1987	Rejected 2001
Clinical outcome	Effective for MDD (Hamilton-D -50%)	Ineffective (no significant improvement)
NeoCortex prediction	Effective (divergence <10%)	High risk (divergence >70%)

4.2.3 Simulation results

Fluoxetine. Starting from a depressed baseline (5HT 29.08%, fatigue 60%), the simulation shows progressive serotonin accumulation over 48 hours. At 8 hours (Tmax) 5HT reaches approximately 35%, a 20% rise from baseline. By 24 hours steady state is approached at 39%. Final measurement at 48 hours: 5HT 42.95% (+13.87 absolute, +47.7% increase), fatigue 51% (an improvement of 9 percentage points), and gut feeling shifts from Neutral to Comfort.

Reboxetine. From a depressed baseline (5HT 27.7%, fatigue 61%), the simulation reveals minimal serotonin change over 48 hours. Despite strong NET inhibition at 90% blockade, noradrenaline increases from 22% to 25% (+14%), but 5HT shows only +2.5% change, from 27.7% to 28.4% — effectively no improvement in the primary therapeutic target for depression. Fatigue improves modestly from 61% to 55%, and gut feeling remains Neutral.

4.2.4 Comparison with published clinical trial data

The clinical endpoints below are measured over weeks with instruments that differ from the simulated readouts. The comparison therefore assesses agreement in direction and relative magnitude, not numerical error.

Fluoxetine clinical evidence

Study / metric	Clinical result	NeoCortex prediction	Agreement
FDA pivotal trials (1985-1987)	Hamilton-D reduction -50% (p<0.001)	Fatigue 60% to 51%	Direction matches (different endpoint)
CSF 5-HIAA	+120-140% at 4-6 weeks	5HT +47.7% at 48h (early phase)	Direction matches; 48 h vs 4-6 weeks
Response rate (>50%)	60-65% respond by week 8	Effective profile (divergence <10%)	Qualitative match
Time to response	Initial improvement 2-4 weeks	Progressive accumulation 24-48h	Early phase only
Side effects	Nausea 21%, insomnia 16%, anxiety 12%	Minimal off-target perturbation	Profile consistent

Sources: FDA Medical Review Documents (1987); Wernicke JF et al. (1987) *J Clin Psychiatry* 48:3-14; Tollefson GD et al. (1993) *Int Psychogeriatr* 8:253-260; Beasley CM et al. (2000) *J Clin Psychiatry* 61:869-877.

Reboxetine clinical evidence (FDA rejection 2001)

Study / metric	Clinical result	NeoCortex prediction	Agreement
FDA review (2001)	Hamilton-D -30%, not superior to placebo	Fatigue 61% to 55%	Direction matches (different endpoint)
CSF 5-HIAA	No significant change (+5%)	5HT +2.5% (no recovery)	Consistent (both near zero)
CSF MHPG	+45-60% increase	NE +14% (22% to 25%)	Directional match
Response rate (>50%)	35-40% (placebo 30%)	Ineffective (divergence >70%)	Consistent
FDA decision	Rejected 2001	High risk (insufficient 5HT recovery)	Matches outcome
European use	Approved in EU, withdrawn UK 2013	Suboptimal profile identified	Consistent with history

Sources: FDA Psychopharmacologic Drugs Advisory Committee Meeting (2001); Eyding D et al. (2010) BMJ 341:c4737; Montgomery S et al. (2003) J Clin Psychopharmacol 23:45-50; Hajós M et al. (2004) CNS Drug Rev 10:23-44.

4.2.5 Comparative analysis summary

Metric	Fluoxetine	Reboxetine	Clinical alignment
Δ Serotonin 48h	+47.7% (29.08% → 42.95%)	+2.5% (27.7% → 28.4%)	Consistent
Δ Noradrenaline	+1.2% (stable)	+14% (22% → 25%)	Correct
Δ Fatigue	-15% (60% → 51%)	-10% (61% → 55%)	Significant vs modest
Δ Gut feeling	+17.3% (Neutral → Comfort)	0% (Neutral)	Matches reports
NeoCortex prediction	Effective (5HT normalisation)	Ineffective (no 5HT recovery)	—
FDA historical outcome	Approved 1987	Rejected 2001	—
Agreement with outcome	Direction matches	Direction matches	2 of 2 correct

4.2.6 Mechanistic interpretation

The simulation reveals why Reboxetine failed. Major Depressive Disorder is primarily a serotonergic disorder: while noradrenaline contributes to symptoms, particularly psychomotor aspects, restoration of serotonin function is necessary for therapeutic response. Reboxetine's high NET selectivity with minimal SERT activity produces noradrenergic activation without addressing the core serotonergic deficit. NeoCortex identifies this mismatch through simulated neurochemistry: the system contains no rule stating that depression requires serotonin. The result follows from the coupling formulas, whose literature-derived coefficients weight serotonin heavily for mood-related readouts, together with the Hill equations and network dynamics.

Clinically the two compounds show distinct profiles. Fluoxetine tends to be more effective in patients with associated anxiety, irritability, somatic symptoms and atypical depression, with nausea, insomnia, agitation and sexual dysfunction as common side effects. Reboxetine proves more useful in patients with anergia, psychomotor slowing and attention deficits, with dry mouth, constipation, tachycardia and insomnia, and a lower incidence of sexual dysfunction.

4.3 Study 2: Epilepsy — Lamotrigine

4.3.1 Pathophysiology

Epilepsy is characterised by recurrent seizures resulting from excessive and hypersynchronous neuronal activity. Under physiological conditions neuronal activity is regulated by a dynamic balance between depolarising and counteracting signals; breaking this balance is the key event in establishing hyperactivity. In epilepsy an increase in excitation or a reduction in inhibition leads to hypersynchronous discharges, the electrophysiological substrate of seizures.

Glutamate, the main excitatory neurotransmitter, shows increased release, altered receptor expression or function, and reduced removal capacity from the synaptic space, promoting prolonged depolarisation and in severe cases excitotoxic damage. GABAergic inhibition may simultaneously be ineffective through reduced synthesis or release, receptor alterations, or ionic gradient modifications that make GABA action less hyperpolarising. Voltage-gated sodium channels enable the rapid repetitive firing that propagates seizures. Neuroimaging demonstrates increased glutamate and reduced GABA in epileptogenic foci.

4.3.2 Lamotrigine: mechanism and clinical efficacy

Lamotrigine is a second-generation antiepileptic acting primarily by voltage-dependent blockade of sodium channels, stabilising them in the inactivated state. This limits the repetitive discharge typical of epileptic activity and reduces presynaptic glutamate release, decreasing postsynaptic excitatory receptor activation and limiting calcium entry. Although it does not directly potentiate GABAergic transmission, it indirectly promotes physiological inhibitory mechanisms. Glutamatergic modulation in cortical and limbic areas also confers mood-stabilising effects, explaining its efficacy in preventing depressive episodes in bipolar disorder. Not acting directly on the GABAergic system, it carries a reduced risk of sedation and cognitive slowing compared with other antiepileptics.

Matsuo et al. (1993, Neurology) showed that lamotrigine at 500 mg/day as add-on therapy reduced median seizure frequency by 36% versus 8% with placebo ($p < 0.05$), with response rates of 33% versus 13%. A meta-analysis by Goa et al. (1993, Drugs) reported reductions of 17-59% across trials, with 13-67% of patients achieving at least 50% seizure reduction. Long-term studies show 33% of children maintaining beneficial response over ten years (Weston et al., 2005, Seizure).

4.3.3 Simulation results

Epilepsy was modelled as a severe excitation-inhibition imbalance with elevated glutamate (65% of normal, representing hyperexcitability) and reduced GABA (36% of normal), giving a baseline E/I ratio of 0.29 against a normal range of 0.10-0.15. Lamotrigine at 200 mg was implemented as glutamate release reduction through sodium channel modulation, indirect GABA facilitation, and membrane stabilisation.

Parameter	Baseline	48h post-treatment	Change	Clinical expectation
Glutamate	65%	53%	-18%	Glu reduction (consistent)
GABA	36%	43%	+19%	GABA increase (consistent)
E/I ratio	0.29 (pathological)	0.10 (normalised)	-66%	E/I normalisation (<0.15 target)
Muscle tension	55%	47%	-8%	Reduced muscle tone
Network stability	Hyperexcitable	Stable	—	Anti-epileptic effect

NeoCortex reproduces the therapeutic mechanism: E/I ratio normalisation from pathological to therapeutic range through emergent glutamate reduction and GABA recovery, outcomes never pre-programmed but arising from sodium channel blockade parameters interacting with network dynamics. Muscle tension reduction aligns with clinical observations of reduced spasticity. The direction of effect agrees with published clinical data.

4.4 Study 3: Parkinson's disease — Levodopa

4.4.1 Pathophysiology

Parkinson's disease is characterised by selective and progressive dopaminergic neurodegeneration in the substantia nigra pars compacta, accompanied by Lewy body formation in surviving neurons. Symptoms typically emerge once 60-80% of nigral dopamine neurons have degenerated, corresponding to a 70-80% reduction in striatal dopamine. This depletion induces marked imbalance between the direct and indirect pathways of the basal ganglia: the indirect pathway is hyperactivated and the direct pathway suppressed, increasing inhibitory output to the thalamus and compromising its ability to modulate cortical activity.

4.4.2 Levodopa: mechanism and clinical efficacy

Levodopa, a dopamine precursor, crosses the blood-brain barrier and is converted to dopamine by residual neurons and glial cells. Its rationale lies not only in neurotransmitter replacement but in the ability to potentiate still-intact motor circuit function: even with substantial neuronal loss it amplifies the residual dopaminergic signal, exploiting the plasticity of surviving networks. Functionally this translates to improvements in bradykinesia, rigidity and movement fluidity. However, the pulsatile and non-physiological nature of the stimulation promotes motor fluctuations and dyskinesias during chronic treatment.

The ELLDOPA trial (Fahn et al., 2004, NEJM) showed dose-dependent UPDRS improvement, with the 600 mg/day group demonstrating sustained benefit two weeks after withdrawal. Pathologically confirmed cases show variable response: 37.5% excellent, 45.7% good, 12.1% moderate and 4.6% poor (Kempster et al., 2019, Mov Disord). The LEAP trial (Verschuur et al., 2019, NEJM) demonstrated that early initiation provides symptomatic relief without modifying disease progression.

4.4.3 Simulation results

Advanced Parkinson's disease was modelled as severe dopamine depletion at 5.7%, representing the 70% nigral neuron loss typical of symptomatic disease. Levodopa at 100 mg was implemented as a boost to dopamine synthesis with gradual accumulation over 48 hours, constrained by the reduced decarboxylase capacity that follows from neuron loss.

Parameter	Baseline	48h post-treatment	Change	Clinical expectation
Dopamine	5.7% (severe depletion)	6.0%	+5.3%	Limited response in advanced PD
Fatigue	56% (exhausted)	50%	-11%	Partial symptom relief
Three-factor learning	Impaired (DA-gated)	Partially restored	—	Incomplete motor learning
Motor function	Severely impaired	Modestly improved	—	Limited by neuron loss

The limited recovery reflects the reality that with 70% neuron loss even precursor supplementation cannot fully restore function, since the enzymatic machinery for conversion is itself depleted. Fatigue improvement matches clinical observations of partial symptomatic relief without disease modification. The system was never programmed with a rule associating advanced disease with poor response: this arises from neuron loss parameters interacting with synthesis dynamics. The direction of effect agrees with published clinical data.

4.5 Cross-validation summary

Three independent retrospective validations across fundamentally different mechanisms support generalisability: serotonergic restoration versus noradrenergic failure in depression; excitation-inhibition normalisation through glutamate and GABA modulation in epilepsy; and limited dopaminergic recovery constrained by neurodegeneration in Parkinson's disease. All three agree in direction with published clinical outcomes using the same NT-physiology coupling framework, suggesting that the 48-coefficient matrix captures neurobiology relevant across pathologies. Quantitative agreement with clinical endpoints over clinically relevant durations remains to be tested.

5. Drug Discovery Module: NEURO-ACh v3.0-OH

5.1 Optimisation methodology

Beyond predicting outcomes of existing drugs, NeoCortex includes an integrated Drug Discovery module combining genetic algorithm optimisation with molecular structure generation. Candidate drugs are represented as vectors of target potencies (SERT, DAT, NET, AChE, MAO-A and others, each in the range 0 to 1). The population evolves through tournament selection, uniform crossover at 70%, and Gaussian mutation of ± 0.15 at 15% rate, with elitism preserving the top five candidates.

Fitness is computed entirely from virtual brain simulation:

$$\text{Fitness} = 0.6 \times \text{NT_Correction} + 0.2 \times (1 - \text{SideEffects}) + 0.1 \times (1 - \text{Complexity}) + 0.1 \times \text{ResponseSpeed}$$

where NT_Correction measures restoration of deficient neurotransmitters to baseline, SideEffects quantifies off-target physiological perturbation, Complexity penalises polypharmacy, and ResponseSpeed rewards rapid therapeutic onset. Each candidate is simulated for 48 hours in the diseased brain, and fitness emerges from actual neurochemical and physiological trajectories rather than from pre-programmed scoring.

5.2 Generated compound

For Alzheimer's disease, modelled as a 60% acetylcholine deficit with 15% serotonin and 10% dopamine deficits, the genetic algorithm converged on a dual-target profile: AChE inhibition 90-95% combined with SERT inhibition 35-45%. This addresses both cognitive symptoms through acetylcholine restoration and neuropsychiatric symptoms through serotonin modulation, a clinically validated strategy.

Property	Value	Comparison
SMILES	<chem>COc1cc2c(cc1OC)C(=O)C(CC2)CC3CCN(CC3)Cc4ccc(F)c(O)c4</chem>	—
Molecular formula	C25H30FNO4	—
Molecular weight	427.52 Da	Similar to donepezil (379.5 Da)
LogP (lipophilicity)	4.60	Improved vs v3.0 (4.89); donepezil 4.36 (same method)
TPSA (polarity)	59.0 Å ²	Better than v3.0 (38.8 Å ²)
H-bond donors	1 (hydroxyl group)	Improved H-bonding
Lipinski violations	0 of 4	Drug-like
QED (drug-likeness)	0.73	Good drug-likeness
SA score (synthesisability)	3.7	Moderately easy synthesis

Calculated properties (RDKit: Crippen LogP, TPSA, QED). The compound is a computational candidate and has not been synthesised or tested.

5.3 Predicted clinical advantages

Cognitive enhancement. Strong AChE inhibition at 90-95%, comparable to donepezil, is expected to improve memory and cognitive function.

Neuropsychiatric benefit. Moderate SERT inhibition at 35-45% addresses depression and anxiety commonly comorbid with Alzheimer's, potentially eliminating the need for a separate SSRI prescription.

Improved tolerability. The meta-hydroxyl group increases polarity (TPSA 59.0 Å² versus 38.8 Å² for v3.0), predicted to reduce gastrointestinal side effects. Clinical AChE inhibitors such as donepezil and rivastigmine exhibit 15-20% incidence of nausea and vomiting; improved polarity may reduce this to 10-15%.

Synthesis feasibility. An SA score of 3.7 indicates moderately easy synthesis, estimated at 6-7 steps from commercially available starting materials, with predicted production cost of roughly \$2,000-3,500/kg, three to five times donepezil.

6. Discussion

6.1 Platform validation and accuracy

NeoCortex shows consistent directional agreement with clinical outcomes across the validation cases. The Fluoxetine versus Reboxetine study provides the clearest evidence: alignment with historical FDA decisions and the correct direction of neurochemical and physiological change. This retrospective result is notable because the system was not trained on these specific outcomes, although the coupling coefficients are derived from the clinical literature. Validation across epilepsy and Parkinson's further supports generalisability, since those outcomes differ fundamentally in mechanism yet are captured by the same coupling framework.

6.2 Emergent versus rule-based systems

A critical distinction: NeoCortex outcomes are emergent, not pre-programmed. The system contains no hard-coded rule stating that depression requires serotonin normalisation, or that fatigue improves when serotonin increases. These principles emerge from disease-specific NT baselines, drug mechanism parameters, biophysical release-reuptake-decay dynamics, the universal coupling coefficients, and network interactions across the simulation. The Reboxetine failure demonstrates this clearly: NET inhibition produces a noradrenaline increase but fails to trigger remission, because the coupling formulas weight serotonin more heavily for fatigue and mood endpoints.

6.3 Drug discovery implications

The generation of NEURO-ACh v3.0-OH showcases utility beyond validation. The genetic algorithm explored the mechanism space to converge on a dual-target profile addressing Alzheimer's dual pathology: cholinergic deficit alongside neuropsychiatric symptoms. The hydroxyl modification, introduced to improve tolerability, exemplifies how emergent fitness functions can guide medicinal chemistry decisions. Traditional docking-based approaches would not capture gastrointestinal tolerability differences between the two variants.

6.4 Limitations and future directions

Current limitations: validation is retrospective and limited to three pathologies, and broader disease coverage is required; molecular docking predictions require experimental validation; genetic variants such as CYP2D6 and COMT polymorphisms are not implemented; drug-drug interactions require multi-drug simulation capability that does not yet exist; and runs are stochastic without a fixed seed, so multi-seed replication with confidence intervals is still to be added.

Long-term exposure remains to be validated. The platform accepts simulated durations from one hour up to a year, and the interface exposes a range of one to fourteen days, but the validation studies reported here used a 48-hour window, chosen for turnaround time during development. The clinical endpoints they are compared against are measured at four to eight weeks, so agreement over longer simulated exposures has not yet been established. This matters in particular for SSRIs, where full therapeutic response takes weeks and is driven by metabolite accumulation.

Development priorities are expansion to bipolar disorder, anxiety and ADHD, to represent a wider variety of pathophysiological phenotypes; integration of neuroimaging data (fMRI, DTI, EEG) for individualised digital twins, enabling patient-level response simulation, adverse-effect risk estimation and optimal dosing within a precision medicine framework; multi-GPU scaling toward 1-10 million neurons, approaching the complexity of biological cortical circuits; interoperability with electronic health records to incorporate longitudinal clinical data and comorbidities; and integration with AI-guided discovery pipelines for systematic exploration of chemical libraries.

6.5 Clinical and economic impact

The potential impact on CNS drug development is substantial. Current Phase II/III failure rates result in \$100-200M losses per failed candidate. NeoCortex could screen out ineffective candidates preclinically, as suggested by the retrospective Reboxetine result; optimise dosing regimens before human trials; identify off-target liabilities early; repurpose existing drugs for new indications; and accelerate personalised medicine through patient-specific simulations. Conservative estimates suggest 40% cost reduction and 2-3 year time savings per successful drug, translating to \$500M-1B savings across a typical pharmaceutical pipeline.

7. Conclusions

NeoCortex is a research platform for virtual clinical trials in CNS drug development, retrospectively validated in direction against clinical outcomes in three pathologies. The four-layer emergent architecture — combining spiking neural networks, complete neuromodulatory systems, advanced plasticity and universal NT-physiology coupling — enables simulation of complex pharmacological effects that traditional methods cannot capture.

Key achievements include discrimination between an FDA-approved and an FDA-rejected antidepressant through simulated neurochemistry; directional validation across three distinct pathologies; a drug discovery demonstration generating NEURO-ACh v3.0-OH, a candidate with a dual-target profile and hypothesised tolerability advantages; and the absence of outcome rules, with results arising from simulated dynamics and literature-derived coupling.

The platform can support pharmaceutical R&D research applications: candidate screening, dose optimisation, mechanism-of-action studies and repurposing initiatives. Future development will add prospective blinded validation, expand disease coverage, integrate patient-specific modelling, and, once validated, pursue regulatory qualification for virtual trial use. NeoCortex suggests that sufficiently detailed neurobiological simulation has the potential to predict clinical outcomes with accuracy sufficient to guide pharmaceutical decision-making — a shift from animal models and static computational approaches to dynamic, emergent, human-relevant virtual trials.

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